## Sub Queries

1. SELECT emp_id, emp_name, dept_id, salary FROM employees e WHERE salary > (SELECT AVG(salary) FROM employees WHERE dept_id = e.dept_id);

2. SELECT emp_id, emp_name, dept_id, salary FROM employees e WHERE salary = (SELECT MAX(salary) FROM employees WHERE dept_id = e.dept_id);

3. SELECT MAX(salary) AS second_highest_salary FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);

4.SELECT emp_name, salary FROM employees WHERE salary = (SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees));

5. SELECT emp_id, emp_name, salary, manager_id FROM employees e WHERE manager_id IS NOT NULL AND salary > (SELECT salary FROM employees m WHERE m.emp_id = e.manager_id);

6.SELECT emp_id, emp_name, salary FROM employees e WHERE EXISTS (SELECT 1 FROM employees r WHERE r.manager_id = e.emp_id AND r.salary > e.salary);

7. SELECT dept_id FROM employees GROUP BY dept_id HAVING AVG(salary) > (SELECT AVG(salary) FROM employees);

8. SELECT emp_id, emp_name, dept_id, salary FROM employees WHERE dept_id IN (SELECT dept_id FROM employees GROUP BY dept_id HAVING AVG(salary) > (SELECT AVG(salary) FROM employees));

9. SELECT emp_id, emp_name, salary FROM employees WHERE salary > ALL (SELECT salary FROM employees WHERE dept_id = (SELECT dept_id FROM departments WHERE dept_name = 'HR'));

10. SELECT emp_id, emp_name, salary FROM employees WHERE salary > ANY (SELECT salary FROM employees WHERE dept_id = (SELECT dept_id FROM departments WHERE dept_name = 'Research'));

11. SELECT dept_id, dept_name FROM departments d WHERE NOT EXISTS (SELECT 1 FROM projects p WHERE p.dept_id = d.dept_id);

12. SELECT emp_id, emp_name, dept_id, salary FROM employees WHERE dept_id IN (SELECT dept_id FROM projects GROUP BY dept_id HAVING COUNT(*) > 1);

13. SELECT emp_id, emp_name, dept_id, salary FROM employees WHERE dept_id IN (SELECT dept_id FROM projects WHERE budget = (SELECT MAX(budget) FROM projects));

14.SELECT emp_id, emp_name, salary FROM employees WHERE ABS(salary - (SELECT AVG(salary) FROM employees)) = (SELECT MIN(ABS(salary - (SELECT AVG(salary) FROM employees))) FROM employees);

15. SELECT DISTINCT e.dept_id FROM employees e WHERE e.salary = (SELECT MAX(salary) FROM employees WHERE dept_id = e.dept_id) AND e.salary > ALL (SELECT m.salary FROM employees m WHERE m.dept_id = e.dept_id AND EXISTS (SELECT 1 FROM employees r WHERE r.manager_id = m.emp_id));